Respiratory System – Page 420

Source: Lucent's General Knowledge – Biology

Biology: Respiratory System

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. The opening of larynx is called: **A.** Pharynx **B.** Glottis **C.** Bronchi **D.** Alveoli	**Ans: B**
Q2. The cartilaginous flap structure that prevents the entry of food during swallowing is: **A.** Larynx **B.** Epiglottis **C.** Trachea **D.** Bronchi	**Ans: B**
Q3. A pair of vocal cords that help in producing sound are present inside the: **A.** Pharynx **B.** Larynx **C.** Trachea **D.** Bronchi	**Ans: B**
Q4. Trachea passes through the neck and from the base of larynx reaches the middle of: **A.** Abdomen **B.** Thoracic cavity (chest) **C.** Cranium **D.** Pelvis	**Ans: B**
Q5. Trachea is supported by: **A.** Bones **B.** C-shaped cartilaginous rings **C.** Muscles only **D.** Ligaments only	**Ans: B**
Q6. The inner lining of trachea is secreted by: **A.** Sweat glands **B.** Ciliated epithelium (mucus secreting) **C.** Bone cells **D.** Nerve cells	**Ans: B**
Q7. Trachea divides into two branches called: **A.** Alveoli **B.** Bronchi **C.** Pharynx **D.** Larynx	**Ans: B**

Q8. Bronchi further divide into smaller branches called: **A.** Alveoli **B.** Bronchioles **C.** Trachea **D.** Larynx	**Ans: B**
Q9. Each lung is surrounded by a membrane called: **A.** Pericardium **B.** Pleural membrane **C.** Peritoneum **D.** Meninges	**Ans: B**
Q10. There is a network of blood capillaries on: **A.** Trachea **B.** Alveoli **C.** Pharynx **D.** Larynx	**Ans: B**
Q11. Each lung has colour: **A.** Red **B.** Pink, spongy **C.** Yellow **D.** Blue	**Ans: B**
Q12. Right lung is divided into: **A.** 2 lobes **B.** 3 lobes **C.** 4 lobes **D.** 5 lobes	**Ans: B**
Q13. Nitrogen percentage in the constitution of air is: **A.** 21% **B.** 79% **C.** 0.03% **D.** 4%	**Ans: B**
Q14. Oxygen percentage in the constitution of air is: **A.** 79% **B.** 21% **C.** 4% **D.** 0.03%	**Ans: B**
Q15. Carbon dioxide percentage in exhaled air is approximately: **A.** 0.03% **B.** 4% **C.** 21% **D.** 79%	**Ans: B**

Q16. The maximum volume of air which can be inhaled through human breathing is: **A.** 200 mL **B.** 400 mL **C.** 600 mL **D.** 800 mL	**Ans: B**

Q17. An adult breathes approximately 12 to 16 times in a: **A.** Hour **B.** Minute **C.** Second **D.** Day	**Ans: B**

Q18. During inspiration, the diaphragm: **A.** Moves up **B.** Flattens (moves down) **C.** Stays same **D.** Disappears	**Ans: B**

Q19. During expiration, air pressure inside and outside the body becomes: **A.** Very different **B.** Equal **C.** Inside is much higher **D.** Outside is zero	**Ans: B**

Q20. The process of respiration can be divided into how many parts? **A.** 2 **B.** 4 parts **C.** 6 **D.** 3	**Ans: B**

Q21. External respiration is the exchange of gases in: **A.** Cells **B.** Lungs **C.** Blood **D.** Stomach	**Ans: B**

Q22. Exchange of gases between blood and tissue fluid is called: **A.** External respiration **B.** Internal respiration **C.** Cellular respiration **D.** Anaerobic respiration	**Ans: B**

Q23. Cellular respiration is where glucose is oxidised by oxygen. This process is called: **A.** Fermentation **B.** Cellular respiration **C.** Photosynthesis **D.** Osmosis	**Ans: B**
Q24. There are how many types of cellular respiration? **A.** 3 **B.** 2 – Anaerobic and Aerobic **C.** 4 **D.** 1	**Ans: B**
Q25. Anaerobic respiration occurs in the absence of: **A.** Water **B.** Oxygen **C.** Carbon dioxide **D.** Nitrogen	**Ans: B**
Q26. During anaerobic respiration, only how many ATP molecules are produced from one molecule of glucose? **A.** 38 **B.** 2 **C.** 4 **D.** 36	**Ans: B**
Q27. In anaerobic respiration in yeast, the final product is: **A.** Lactic acid **B.** Ethyl alcohol and CO2 **C.** Water **D.** Oxygen	**Ans: B**
Q28. In anaerobic respiration in animals, the final product is: **A.** Ethyl alcohol **B.** Lactic acid **C.** Acetic acid **D.** Ethanol	**Ans: B**
Q29. A sprint athlete feels cramps due to accumulation of: **A.** Glucose **B.** Lactic acid **C.** Ethanol **D.** CO2	**Ans: B**

Q30. Aerobic respiration takes place in presence of oxygen. The complete process produces how many ATP? **A.** 2 **B.** 38 **C.** 4 **D.** 36	**Ans: B**

Q31. Glycolysis (cytoplasm) was first studied by: **A.** Kreb **B.** Embden, Meyerhof, Parnas (EMP path) **C.** Watson **D.** Mendel	**Ans: B**

Q32. Glycolysis is present in both types of respiration and takes place in: **A.** Mitochondria **B.** Cytoplasm **C.** Nucleus **D.** Ribosome	**Ans: B**

Q33. Kreb's Cycle was described by: **A.** Embden **B.** Hans Krebs in 1937 **C.** Watson in 1953 **D.** Mendel in 1865	**Ans: B**

Q34. Kreb's Cycle is completed inside the: **A.** Cytoplasm **B.** Mitochondria **C.** Nucleus **D.** Ribosome	**Ans: B**

Q35. Kreb's cycle is also called: **A.** EMP pathway **B.** Citric Acid Cycle or TCA cycle **C.** Calvin cycle **D.** Nitrogen cycle	**Ans: B**

Q36. Breakdown of pyruvate into CO_2, H_2O and energy takes place in: **A.** Cytoplasm **B.** Mitochondria **C.** Nucleus **D.** Cell membrane	**Ans: B**

Q37. In glycolysis, 4 atoms of hydrogen formed are used in converting NAD to: **A.** ATP **B.** 2NADH2 **C.** FAD **D.** GTP	**Ans: B**
Q38. The enzymes which take part in glycolysis and respiration are found in: **A.** Nucleus **B.** Cell cytoplasm **C.** Ribosome **D.** Golgi body	**Ans: B**
Q39. To start glycolysis, how many molecules of ATP are utilized? **A.** 4 **B.** 2 **C.** 1 **D.** 6	**Ans: B**
Q40. As a result of glycolysis, how many ATP are obtained net? **A.** 4 **B.** 2 (net, since 4 produced minus 2 used) **C.** 38 **D.** 36	**Ans: B**
Q41. There is no need of oxygen in: **A.** Aerobic respiration **B.** Glycolysis **C.** Kreb's cycle **D.** Electron transport chain	**Ans: B**
Q42. Breaking down of glucose in cytoplasm results in formation of: **A.** Ethanol **B.** Pyruvate (pyruvic acid) **C.** Lactic acid **D.** Acetic acid	**Ans: B**
Q43. ATP stands for: **A.** Adenine Tri Phosphate **B.** Adenosine Triphosphate **C.** Amino Tri Phosphate **D.** Adenosine Tri Protein	**Ans: B**
Q44. Transportation of oxygen from lungs to the cells of body takes place by: **A.** White blood cells **B.** Haemoglobin (in RBC) **C.** Plasma only **D.** Platelets	**Ans: B**

Q45. Carbon dioxide from cells to lungs is mainly transported as: **A.** Attached to hemoglobin **B.** Bicarbonates (70% as HCO3) **C.** Dissolved in plasma only **D.** In WBCs	**Ans: B**
Q46. Carbon dioxide after mixing with plasma forms: **A.** Oxygen **B.** Carbonic acid **C.** Lactic acid **D.** Acetic acid	**Ans: B**
Q47. Respiration is controlled by: **A.** Cerebrum **B.** Medulla in second (hindbrain) **C.** Cerebellum **D.** Hypothalamus	**Ans: B**
Q48. Respiration is a: **A.** Anabolic process **B.** Catabolic process (it also reduces the weight of the body) **C.** Neither **D.** Synthetic process	**Ans: B**
Q49. Cyanide poisoning causes death by: **A.** Stopping digestion **B.** Breaking down the electron transport chain system **C.** Stopping heartbeat **D.** Blocking kidneys	**Ans: B**
Q50. The gaseous exchange in lungs is called: **A.** Internal respiration **B.** External respiration **C.** Cellular respiration **D.** Anaerobic respiration	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — Glottis
Page 420: The opening of larynx is called glottis.

Topic: Respiratory Anatomy

Q2. Answer: B — Epiglottis
Page 420: Epiglottis is the cartilaginous flap that prevents food entry during swallowing.

Topic: Respiratory Anatomy

Q3. Answer: B — Larynx
Page 420: A pair of vocal cords are present inside the larynx.

Topic: Respiratory Anatomy

Q4. Answer: B — Thoracic cavity (chest)
Page 420: Trachea from the base of larynx reaches the middle of thoracic cavity.

Topic: Respiratory Anatomy

Q5. Answer: B — C-shaped cartilaginous rings
Page 420: Trachea is called wind pipe, supported by C-shaped cartilaginous rings.

Topic: Respiratory Anatomy

Q6. Answer: B — Ciliated epithelium (mucus secreting)
Page 420: The inner lining of trachea is ciliated epithelium (mucus secreting).

Topic: Respiratory Anatomy

Q7. Answer: B — Bronchi
Page 420: Trachea divides into two branches called bronchi, each entering one lung.

Topic: Respiratory Anatomy

Q8. Answer: B — Bronchioles
Page 420: Bronchi further divide into smaller branches called bronchioles, finally giving alveoli.

Topic: Respiratory Anatomy

Q9. Answer: B — Pleural membrane
Page 420: Each lung is surrounded by a membrane called pleural membrane.

Topic: Respiratory Anatomy

Q10. Answer: B — Alveoli
Page 420: There is a network of blood capillaries on alveoli.

Topic: Respiratory Anatomy

Q11. Answer: B — Pink, spongy
Page 420: Each lung is pink, spongy in comparison to left lung.

Topic: Respiratory Anatomy

Q12. Answer: B — 3 lobes
Page 420: Right lung has three lobes and left lung has two lobes.

Topic: Respiratory Anatomy

Q13. Answer: B — 79%
Page 420: Constitution of air: Nitrogen 79%, Oxygen 21%, Carbon dioxide 0.03%.

Topic: Respiration

Q14. Answer: B — 21%
Page 420: Constitution of air: Oxygen 21%.

Topic: Respiration

Q15. Answer: B — 4%
Page 420: In exhaled air, CO_2 is about 4% (vs 0.03% in inhaled air).

Topic: Respiration

Q16. Answer: B — 400 mL
Page 420: Maximum volume of air inhaled is approximately 400 mL.

Topic: Respiration

Q17. Answer: B — Minute
Page 420: An adult breathes approximately 12 to 16 times in a minute.

Topic: Respiration

Q18. Answer: B — Flattens (moves down)
Page 420: During inspiration, the diaphragm flattens/contracts, creating low pressure, allowing air to enter lungs.

Topic: Respiration

Q19. Answer: B — Equal
Page 420: In expiration, diaphragm relaxes, air pressure inside and outside body becomes equal.

Topic: Respiration

Q20. Answer: B — 4 parts
Page 420: Process of respiration is divided into 4 parts: External, Internal, Transportation of gases, Cellular respiration.

Topic: Respiration

Q21. Answer: B — Lungs
Page 420: External respiration is exchange of gases in lungs.

Topic: Respiration

Q22. Answer: B — Internal respiration
Page 420: Internal respiration is exchange of gases between blood and tissue fluid.

Topic: Respiration

Q23. Answer: B — Cellular respiration

Page 420: Cellular respiration: glucose is oxidised by oxygen, reached into the cell.

Topic: Respiration

Q24. Answer: B — 2 – Anaerobic and Aerobic

Page 420: Two types of cellular respiration: Anaerobic and Aerobic.

Topic: Respiration

Q25. Answer: B — Oxygen

Page 420: Anaerobic respiration is if the oxidation of food takes place in absence of oxygen.

Topic: Respiration

Q26. Answer: B — 2

Page 420: During anaerobic respiration, only 2 ATP molecules are produced.

Topic: Respiration

Q27. Answer: B — Ethyl alcohol and CO_2

Page 420: In yeast: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ + Energy (ethyl alcohol or ethanol).

Topic: Respiration

Q28. Answer: B — Lactic acid

Page 420: In animals: anaerobic respiration produces lactic acid. Accumulation causes muscle pain (cramps).

Topic: Respiration

Q29. Answer: B — Lactic acid

Page 420: After a sprint, lactic acid accumulates causing pain in the thigh muscles.

Topic: Respiration

Q30. Answer: B — 38

Page 420: Complete aerobic respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + 2870 kJ energy (38 ATP).

Topic: Respiration

Q31. Answer: B — Embden, Meyerhof, Parnas (EMP path)

Page 420: Glycolysis was first studied by Embden, Meyerhof, Parnas. Also called EMP pathway.

Topic: Respiration

Q32. Answer: B — Cytoplasm

Page 420: Glycolysis is present in both aerobic and anaerobic respiration, takes place in cytoplasm.

Topic: Respiration

Q33. Answer: B — Hans Krebs in 1937

Page 420: Kreb's Cycle was described by Hans Krebs in 1937. Also called Citric Acid Cycle or TCA cycle.

Topic: Respiration

Q34. Answer: B — Mitochondria

Page 420: Kreb's cycle is completed inside the Mitochondria in presence of specific enzymes.

Topic: Respiration

Q35. Answer: B — Citric Acid Cycle or TCA cycle

Page 420: Kreb's cycle is also called Citric Acid Cycle or Tricarboxylic Acid Cycle (TCA cycle).

Topic: Respiration

Q36. Answer: B — Mitochondria

Page 420: Breakdown of pyruvate takes place in mitochondria.

Topic: Respiration

Q37. Answer: B — 2NADH2

Page 420: 4 atoms of hydrogen formed in glycolysis are used in converting NAD to 2NADH2.

Topic: Respiration

Q38. Answer: B — Cell cytoplasm

Page 420: Enzymes for glycolysis are found in cell cytoplasm.

Topic: Respiration

Q39. Answer: B — 2

Page 420: To start the process of glycolysis, 2 molecules of ATP are utilised.

Topic: Respiration

Q40. Answer: B — 2 (net, since 4 produced minus 2 used)

Page 420: As a result of glycolysis, 4 molecules of ATP are obtained. But 2 were used, so net gain is 2 ATP.

Topic: Respiration

Q41. Answer: B — Glycolysis

Page 420: There is no need of oxygen in glycolysis. This process is similar in both anaerobic and aerobic respiration.

Topic: Respiration

Q42. Answer: B — Pyruvate (pyruvic acid)

Page 420: Breaking down of glucose in cytoplasm results in formation of two molecules of pyruvate and energy.

Topic: Respiration

Q43. Answer: B — Adenosine Triphosphate

Page 420: ATP = Adenosine Triphosphate.

Topic: Respiration

Q44. Answer: B — Haemoglobin (in RBC)

Page 420: Transportation of oxygen takes place by haemoglobin in blood.

Topic: Respiration

Q45. Answer: B — Bicarbonates (70% as HCO3)

Page 420: 70% of CO_2 is transported as bicarbonates, rest by mixing with plasma and combining with haemoglobin.

Topic: Respiration

Q46. Answer: B — Carbonic acid

Page 420: Carbon dioxide after mixing with plasma forms carbonic acid.

Topic: Respiration

Q47. Answer: B — Medulla in second (hindbrain)

Page 420: Respiration is controlled by medulla in the hindbrain.

Topic: Respiration

Q48. Answer: B — Catabolic process (it also reduces the weight of the body)

Page 420: Respiration is a catabolic process. It also reduces the weight of the body.

Topic: Respiration

Q49. Answer: B — Breaking down the electron transport chain system

Page 420: Cyanide poisoning causes death by breaking down the electron transport chain system.

Topic: Respiration

Q50. Answer: B — External respiration

Page 420: The gaseous exchange in lungs is called external respiration.

Topic: Respiration